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Wang et al.

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(54) **CONDENSING HEAT EXCHANGER AND WATER HEATER HAVING SAME**

(58) **Field of Classification Search**

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(57) **ABSTRACT**

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(51) **Int. Cl.**

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F24H 1/00 (2022.01)

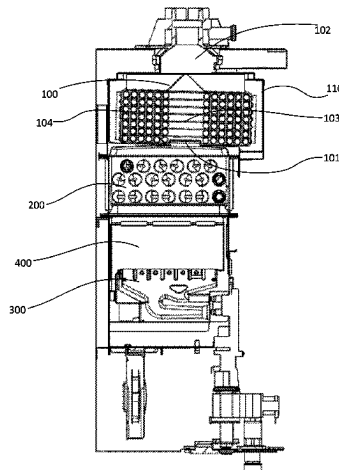
F24H 8/00 (2022.01)

(52) **U.S. Cl.**

CPC **F24H 8/006** (2013.01); **F24H 1/0036** (2013.01); **F24H 9/0026** (2013.01)

A condensing heat exchanger (100) capable of lowering a requirement for a fan and a water heater having the same, wherein the condensing heat exchanger (100) comprises: a heat exchanger housing (110) having a flue gas inlet (101) and a flue gas outlet (102); and a heat exchange unit (104) located within the heat exchanger housing (110), wherein a sidewall flue gas exhaust channel (105) communicated with the flue gas outlet (102) is provided between the heat exchange unit (104) and an inner sidewall of the heat exchanger housing (110), the heat exchange unit (104) surrounds a flue gas inlet channel (103) communicated with the flue gas inlet (101), the sidewall flue gas exhaust channel

(Continued)



(105) surrounds the heat exchange unit (104), and the heat exchange unit (104) is internally provided with a heat exchange flue which communicates the flue gas inlet channel (103) with the sidewall flue gas exhaust channel (105).

21 Claims, 6 Drawing Sheets

(58) **Field of Classification Search**

CPC F28D 7/16; F28F 3/12; F28F 3/14; F28B 9/08

See application file for complete search history.

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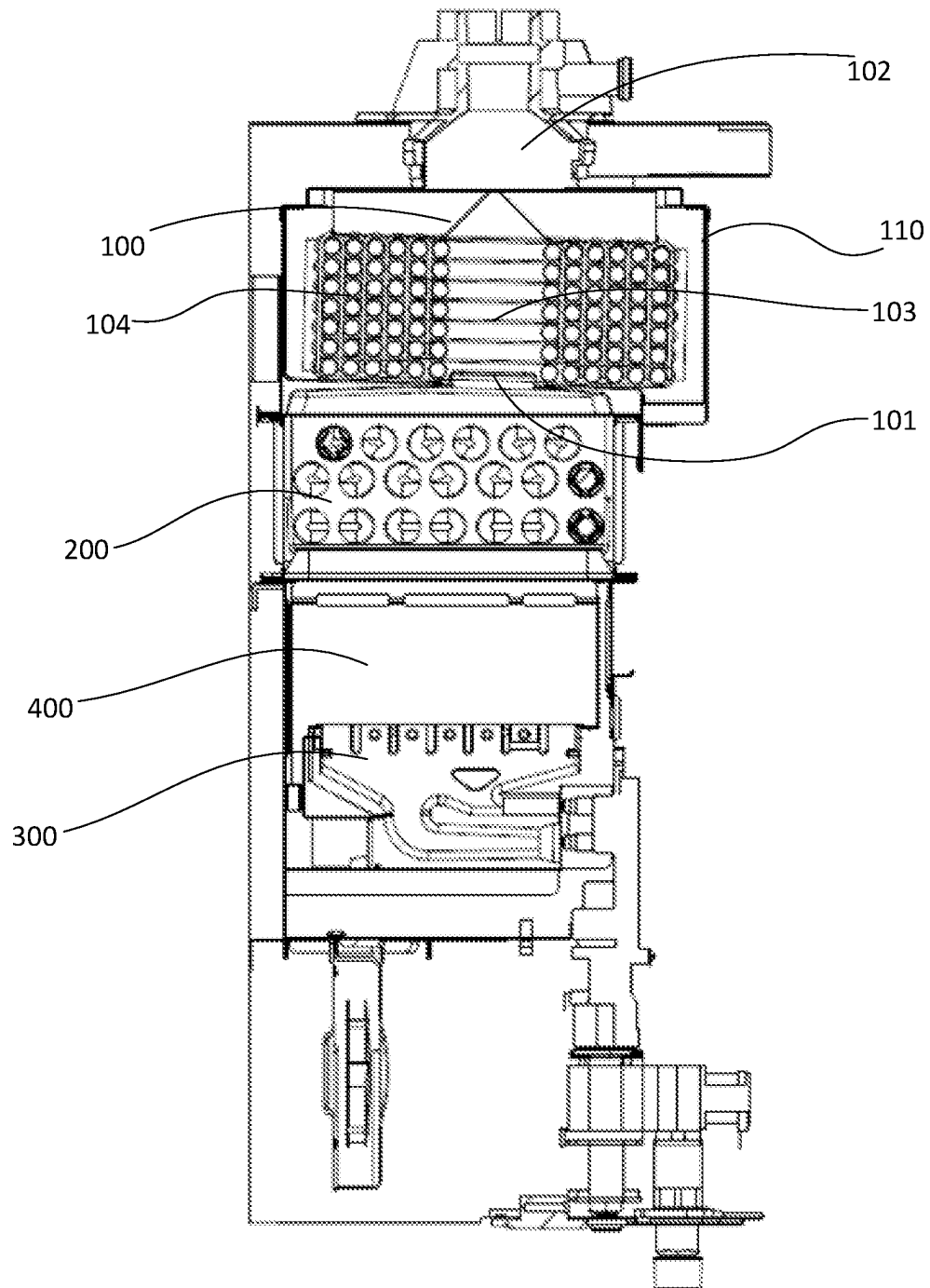


FIG. 1

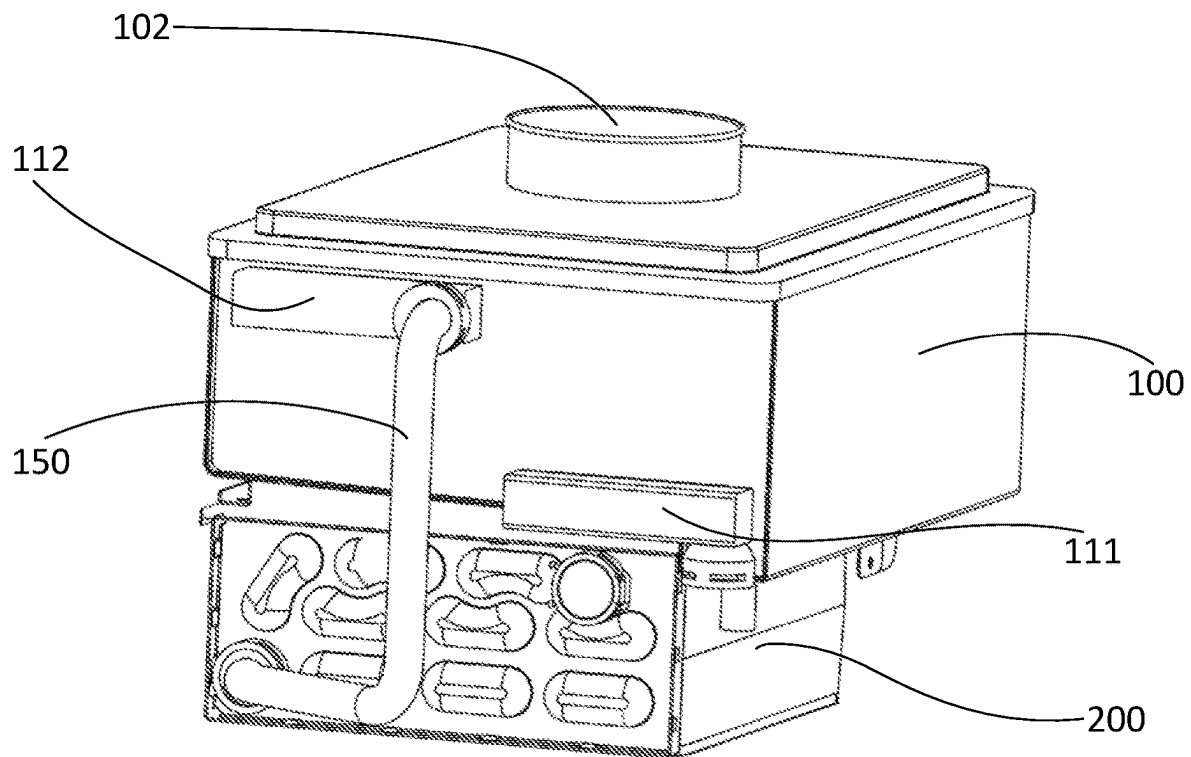


FIG. 2

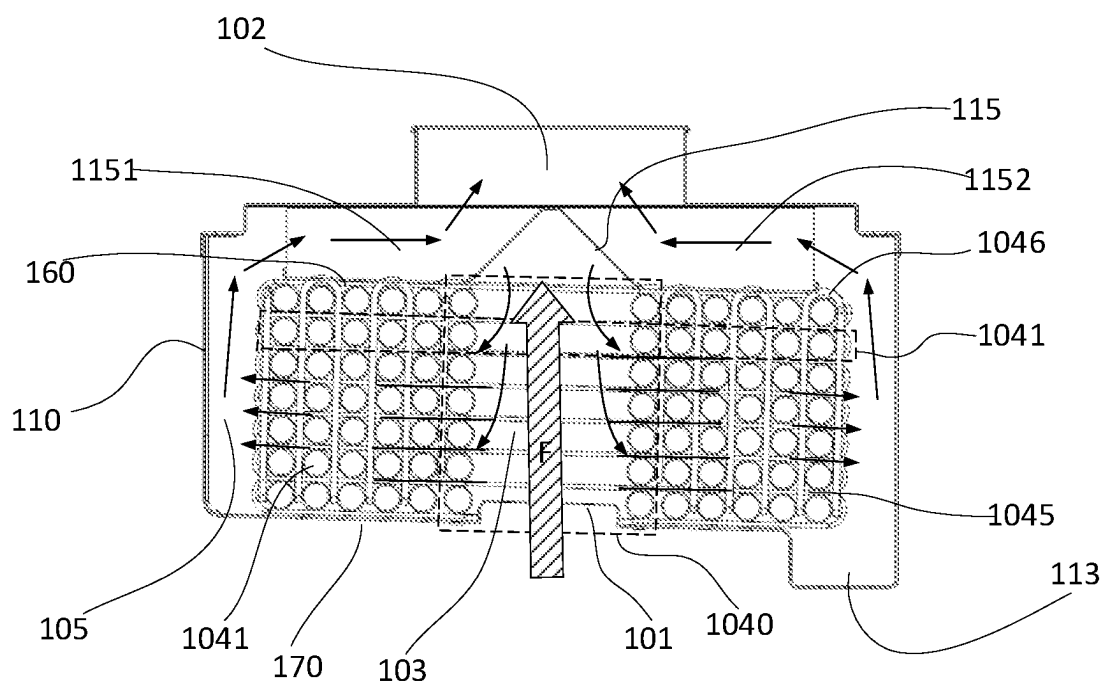


FIG. 3

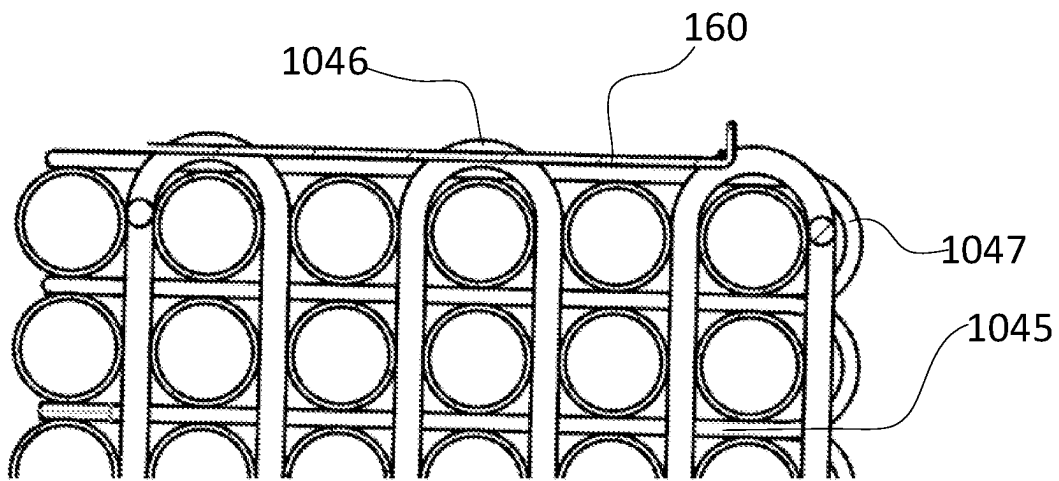


FIG. 4

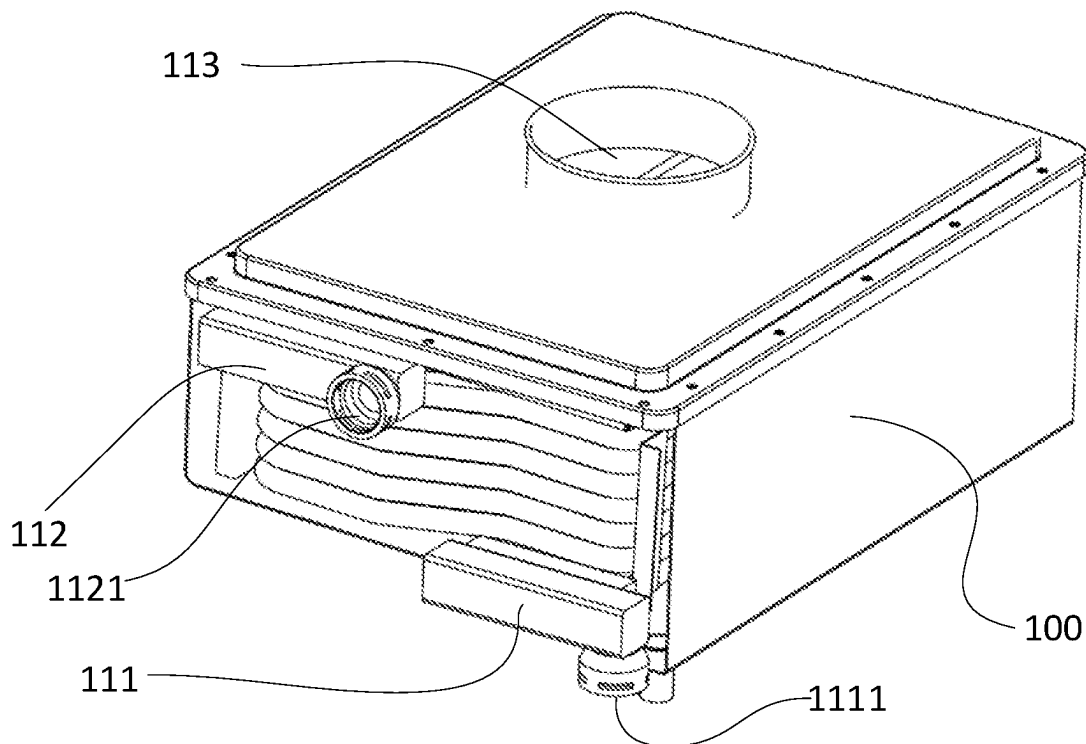


FIG. 5

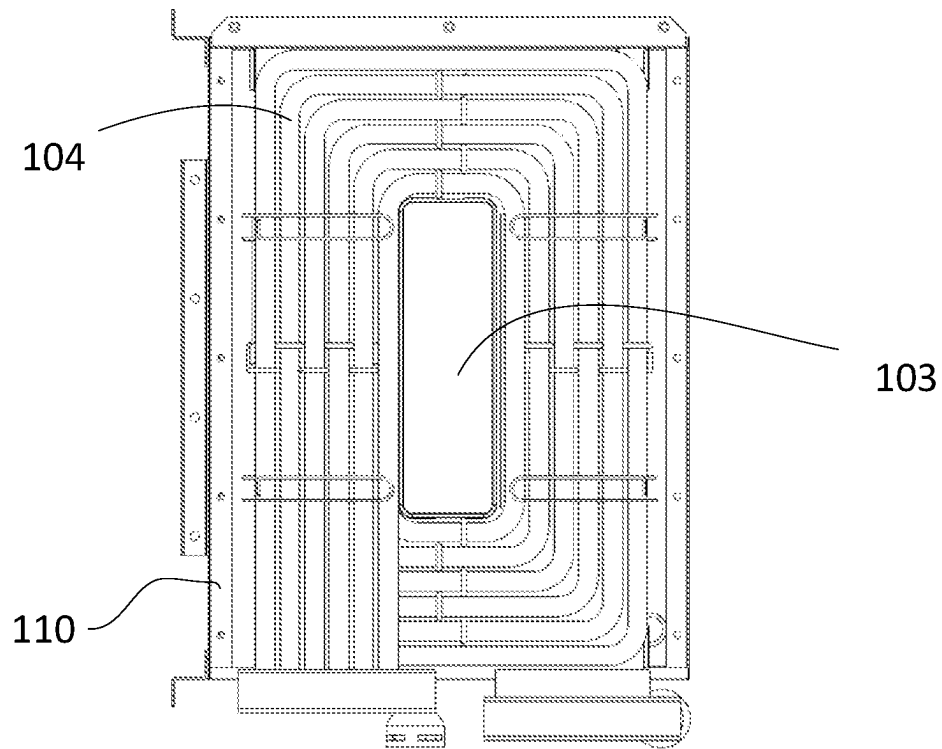


FIG. 6

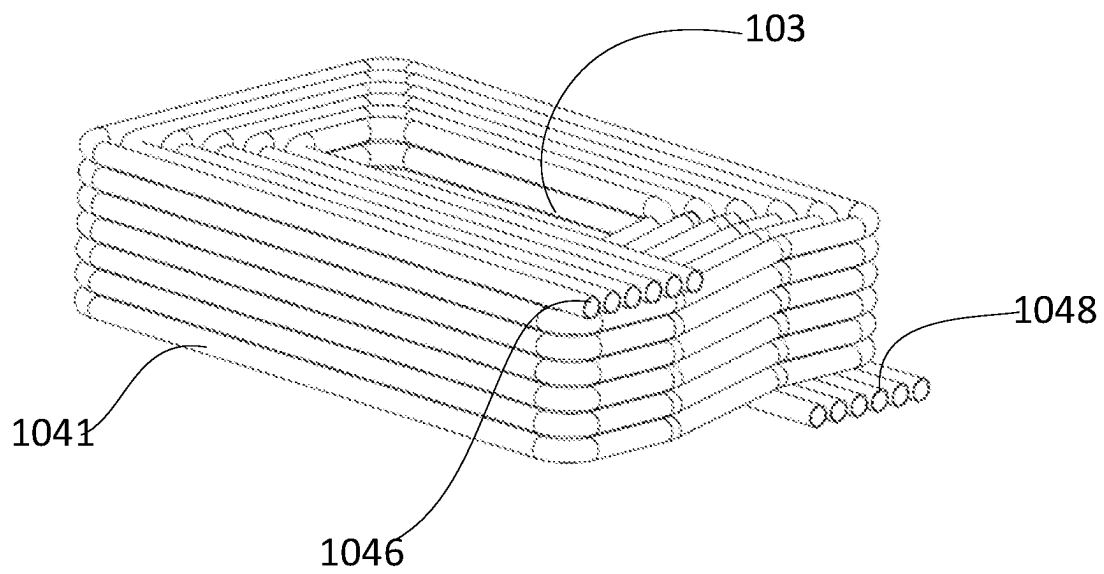


FIG. 7

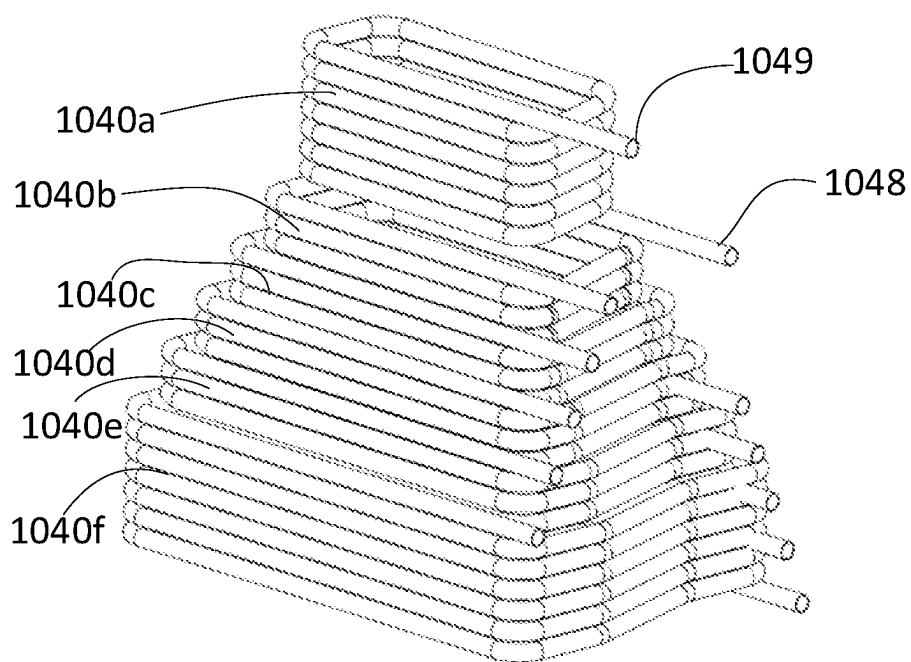


FIG. 8

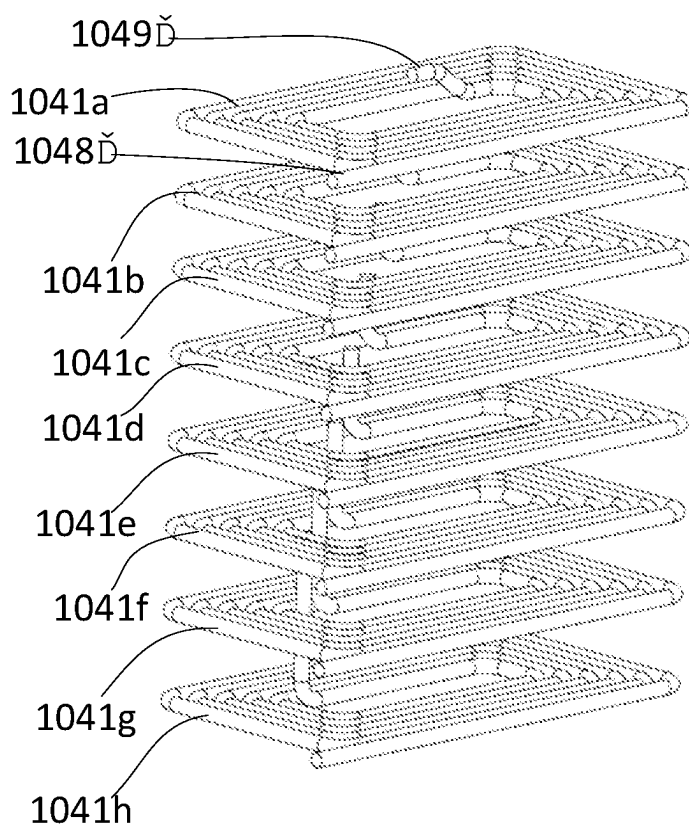


FIG. 9

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**CONDENSING HEAT EXCHANGER AND
WATER HEATER HAVING SAME****CROSS-REFERENCE TO RELATED
APPLICATION(S)**

This application is a national phase entry of PCT/CN2021/114193, filed on Aug. 24, 2021, which claims priority to CN 202011417846.7, filed on Dec. 7, 2020, the entire contents of which are fully incorporated herein by reference.

RELATED APPLICATION

This application claims priority of the Chinese patent application No. 202011417846.7 filed on Dec. 7, 2020, which is incorporated herein by reference.

FIELD

The present disclosure relates to the field of water heating, and particularly to a condensing heat exchanger and a water heater having the same.

BACKGROUND

At present, gas water heaters have been widely used, in which a condensing gas water heater, as an efficient gas combustion device, has a great energy saving potential. Compared with the ordinary gas water heaters, the condensing gas water heater is further provided with a condensing heat exchanger above a sensible heat exchanger, which can fully absorb the heat of high-temperature flue gas and improve the heat exchange efficiency of the whole machine.

However, the existing condensing gas water heater is complicated in the design of the flue gas circulation path, which leads to a large resistance to flue gas flow and a high requirement for a fan. Moreover, the condensed water is not easy to be discharged, and the condensing efficiency is low.

SUMMARY

In view of the above problems, an objective of the present disclosure is to provide a condensing heat exchanger capable of lowering a requirement for a fan and a water heater having the same.

Another objective of the present disclosure is to provide a condensing heat exchanger and a water heater having the same, so as to improve a condensing efficiency.

In order to achieve at least one of the above objectives, the present disclosure adopts the following technical solution:

A condensing heat exchanger, comprising:
a heat exchanger housing having a flue gas inlet and a flue gas outlet; and
a heat exchange unit located in the heat exchanger housing,

wherein a sidewall flue gas exhaust channel communicated with the flue gas outlet is provided between the heat exchange unit and an inner sidewall of the heat exchanger housing, the heat exchange unit surrounds a flue gas inlet channel communicated with the flue gas inlet, the sidewall flue gas exhaust channel surrounds the heat exchange unit, and the heat exchange unit is internally provided with a heat exchange flue which communicates the flue gas inlet channel with the sidewall flue gas exhaust channel.

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As a preferred embodiment, an end of the heat exchange unit close to the flue gas outlet is shielded and blocked by a shielding plate, and the flue gas inlet channel extends from the flue gas inlet to the shielding plate.

As a preferred embodiment, the heat exchange unit comprises a plurality of heat exchange tube layers stacked in a vertical direction, the heat exchange tube layers comprise a plurality of heat exchange tube rings nested with one another and having surrounded areas gradually decreasing from outside to inside in a radial direction, and the heat exchange flue comprises a spacing flue located between adjacent two of the heat exchange tube layers in an up-down direction.

As a preferred embodiment, the heat exchange tube rings having the same surrounded area of the different heat exchange tube layers are stacked in the vertical direction to form a tube ring stack.

As a preferred embodiment, the tube ring stack is formed by one heat exchange tube spirally extending from the water inlet to the water outlet, and the heat exchange tube layer is formed by the heat exchange tube rings located at the same height of the different tube ring stacks.

As a preferred embodiment, the heat exchanger housing is provided with a water inlet part and a water outlet part, and at least two of the tube ring stacks are connected in parallel between the water inlet part and the water outlet part.

As a preferred embodiment, the heat exchanger housing is provided with a water inlet part and a water outlet part, and at least two of the tube ring stacks are connected in series between the water inlet and the water outlet.

As a preferred embodiment, the heat exchange unit comprises a first number of tube ring stacks connected in parallel and a second number of tube ring stacks connected in parallel, and the first number of tube ring stacks and the second number of tube ring stacks are connected in series between the water inlet part and the water outlet part.

As a preferred embodiment, the heat exchange tube layer is formed by one heat exchange tube spirally winding from outside to inside.

As a preferred embodiment, at least two of the heat exchange tube layers are connected in parallel between the water inlet part and the water outlet part, and/or at least two of the heat exchange tubes are connected in series between the water inlet part and the water outlet part.

As a preferred embodiment, the heat exchanger housing is provided with a water inlet part and a water outlet part, and the heat exchange unit comprises a heat exchange tube communicated between the water inlet part and the water outlet part.

As a preferred embodiment, the water inlet part is disposed close to a bottom of the heat exchanger housing, and the water outlet part is disposed close to a top of the heat exchanger housing.

As a preferred embodiment, a vertical spacing between adjacent two of the heat exchange tube layers is less than a horizontal spacing between adjacent two of the tube ring stacks.

As a preferred embodiment, a vertical spacing between adjacent two of the heat exchange tube layers is greater than 0.68 mm and less than 2.66 mm.

As a preferred embodiment, the heat exchange unit is further provided with a spacer bar for spacing adjacent two of the heat exchange tube layers.

As a preferred embodiment, the heat exchange unit is further provided with a spacer bar for spacing adjacent two of the tube ring stacks and attached thereto.

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As a preferred embodiment, a bottom of the heat exchanger housing is further provided with a condensed water outlet located at a lowest position at the bottom of the heat exchanger housing.

As a preferred embodiment, the bottom of the heat exchanger housing has a bearing surface for bearing the heat exchange unit and wholly inclined to the condensed water outlet from a side away from the condensed water outlet.

As a preferred embodiment, a height of the heat exchange unit or at least one of the heat exchange tube layers gradually decreases from a side away from the condensed water outlet to a side close to the condensed water outlet.

As a preferred embodiment, a bottom plate of the heat exchanger housing forms the bearing surface by stamping.

As a preferred embodiment, an inclination angle of the bearing surface with respect to a horizontal plane is within 10 degrees.

As a preferred embodiment, a spacing protrusion is disposed between the shielding plate and a top wall of the heat exchanger housing to partition an area above the shielding plate to form top flow channels communicated with the flue gas outlet, respectively.

A water heater, comprising:

a primary heat exchanger; and

the condensing heat exchanger according to any of the above embodiments and located above the primary heat exchanger.

Advantageous Effects

In the condensing heat exchanger according to the embodiment of the present disclosure, the flue gas inlet channel is surrounded by the heat exchange unit and located at a center thereof, such that flue gas enters the center of the heat exchange unit via the flue gas inlet channel, diffuses around through the heat exchange unit to enter the sidewall flue gas exhaust channel, and then is discharged from the flue gas outlet, thereby reducing the flue gas resistance and then lowering the requirement for the fan.

With reference to the following descriptions and drawings, the particular embodiments of the present disclosure will be disclosed in detail to indicate the ways in which the principle of the present disclosure can be adopted. It should be understood that the scope of the present disclosure is not limited to the embodiments.

The features described and/or illustrated with respect to one embodiment may be used in one or more other embodiments in the same or similar way, may be combined with the features in other embodiments, or may take place of those features.

It should be emphasized that the term 'comprise/include' as used herein refers to the presence of features, integers, steps or components, but does not exclude the presence or addition of one or more other features, integers, steps or components.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings described herein are used only for the purpose of illustration and are not intended to limit the scope of the present disclosure in any way. In addition, the shapes, the scales, and the like of the components in the drawings are merely schematic so as to facilitate the understanding of the present disclosure, rather than specific limitations thereof. Under the teaching of the present disclosure, those skilled in the art can select various possible shapes and scales based on specific conditions to implement the present disclosure.

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FIG. 1 is a partial view of a water heater according to an embodiment of the present disclosure;

FIG. 2 is a view of an assembly of a condensing heat exchanger and a primary heat exchanger of FIG. 1;

FIG. 3 is a cross-sectional view of a condensing heat exchanger of FIG. 2;

FIG. 4 is a partial enlarged view of FIG. 3;

FIG. 5 is a schematic view of a condensing heat exchanger of FIG. 2;

FIG. 6 is an internal top view of FIG. 5;

FIG. 7 is a schematic view of a heat exchange unit of FIG. 5;

FIG. 8 is an exploded view of FIG. 7;

FIG. 9 is a schematic view of a heat exchange unit according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

In order to enable those skilled in the art to better understand the technical solutions of the present disclosure, the technical solutions of the embodiments of the present disclosure will be clearly and completely described below in conjunction with the drawings of the embodiments of the present disclosure. Obviously, those embodiments described are only a part, rather than all, of the embodiments of the present disclosure. Based on the embodiments of the present disclosure, any other embodiment obtained by those of ordinary skill in the art without paying any creative effort should fall within the protection scope of the present disclosure.

It should be noted that when an element is referred to as being 'disposed' on another element, it may be directly on another element or there may be an intermediate element. When an element is regarded as being 'connected' to another element, it may be directly connected to another element or there may be an intermediate element at the same time. The terms 'vertical', 'horizontal', 'left', 'right', and similar expressions used herein are only for the purpose of illustration, rather than indicating a unique implementation.

Unless otherwise defined, all technical and scientific terms used herein have the same meanings as those commonly understood by those skilled in the art of the present disclosure. The terms used herein in the specification of the present disclosure are only for the purpose of describing the specific embodiments, rather than limiting the present disclosure. The term 'and/or' used herein includes any and all combinations of one or more of the related items listed.

Referring to FIGS. 1 to 8, an embodiment of the present disclosure provides a condensing heat exchanger 100, which includes: a heat exchanger housing 110 having a flue gas inlet 101 and a flue gas outlet 102; and a heat exchange unit 104 located within the heat exchanger housing 110, wherein a sidewall flue gas exhaust channel 105 communicated with the flue gas outlet 102 is provided between the heat exchange unit 104 and an inner sidewall of the heat exchanger housing 110. The heat exchange unit 104 surrounds a flue gas inlet channel 103 communicated with the flue gas inlet 101; the sidewall flue gas exhaust channel 105 surrounds the heat exchange unit 104; the heat exchange unit 104 is internally provided with a heat exchange flue which communicates the flue gas inlet channel 103 with the sidewall flue gas exhaust channel 105.

In this embodiment, the heat exchanger housing 110 is a cuboid structure as a whole, with an accommodation space for accommodating the heat exchange unit 104 formed therein. The peripheral sidewalls of the heat exchanger housing 110 are two pairs of opposite sidewalls, and corre-

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spondingly, the heat exchange unit **104** therein is substantially a cuboid structure as a whole which is matched with the shape of the heat exchanger housing **110**. The heat exchanger housing **110** has opposite first and second sidewalls and opposite third and fourth sidewalls, wherein the first, third, second, and fourth sidewalls are annularly arranged in sequence.

As illustrated in FIGS. **5** and **6**, with the first and second sidewalls being taken as left and right sidewalls, and the third and fourth sidewalls being taken as front and rear sidewalls, left and right sidewall flue gas exhaust channels **105** are formed between the first and second sidewalls and the peripheral sidewalls of the heat exchange unit **104**, respectively, and front and rear sidewall flue gas exhaust channels **105** are formed between the third and fourth sidewalls and the peripheral sidewalls of the heat exchange unit **104**, respectively. The left sidewall flue gas exhaust channel **105**, the rear sidewall flue gas exhaust channel **105**, the right sidewall flue gas exhaust channels **105**, and the front sidewall flue gas exhaust channel **105** are sequentially communicated to form an annular sidewall flue gas exhaust channel **105** to surround the heat exchange unit **104**.

In the condensing heat exchanger **100** according to this embodiment, the flue gas inlet channel **103** is surrounded by the heat exchange unit **104** and located at a center thereof, such that flue gas enters the center of the heat exchange unit **104** via the flue gas inlet channel **103**, diffuses around through the heat exchange unit **104** to enter the sidewall flue gas exhaust channel **105**, and then is discharged from the flue gas outlet **102**, thereby reducing the flue gas resistance and then lowering a requirement for a fan.

When passing through the heat exchange unit **104** via the heat exchange flues inside the heat exchange unit **104**, the flue gas fully exchanges heat with the heat exchange unit **104**. After the condensing heat exchanger **100** according to this embodiment exchanges heat with the flue gas at the heat exchange efficiency of above 0.92, the temperature of the discharged flue gas is about 21° C. With the same primary heat exchanger **200**, the condensing heat exchanger **100** according to this embodiment achieves a heat exchange efficiency equal to or greater than that of the traditional condensing heat exchanger **100** with a lateral flue gas intake (in the experiment, the outlet flue gas temperature is reduced by about 4° C. compared with the traditional example), the flue gas resistance is lower, and a fan speed is reduced by 500 revolutions under a maximum load, so that a low-pressure fan can meet the heat exchange requirement of the water heater.

The condensing heat exchanger **100** may serve as a secondary heat exchanger cooperated with the primary heat exchanger **200** (e.g., a finned tube heat exchanger) to exchange heat with the flue gas so as to heat water. The condensing heat exchanger **100** may be mounted above the primary heat exchanger **200** to continuously exchange heat with the flue gas flowing therefrom. The flue gas enters the condensing heat exchanger **100** from a middle portion thereof, and flows around to exchange heat with the heat exchange tubes of the heat exchange unit **104**, thereby reducing the flue gas flow resistance and lowering the requirement for the fan.

In this embodiment, the heat exchange unit **104** includes a plurality of turns of heat exchange tubes (also called as heat exchange coils) surrounding the flue gas inlet channel **103**. The heat exchange tube may be a corrugated tube. As illustrated in FIGS. **1** and **2**, the heat exchanger housing **110** is provided with a water inlet part **111** and a water outlet part **112**. The water inlet part **111** is configured to input water

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(e.g., cold water) into the heat exchange unit **104**, and the water outlet part **112** is configured to output water after the heat exchange between the heat exchange unit **104** and the flue gas. In which, the water inlet part **111** may be communicated with a cold-water input end of the water heater to input cold water. The water outlet part **112** may be communicated with the primary heat exchanger **200**, so that the water flowing out of the condensing heat exchanger **100** can be continuously heated by the primary heat exchanger **200**. The primary heat exchanger **200** may be a finned tube heat exchanger. A combustor **300** is disposed below the primary heat exchanger **200**, with a combustion chamber **400** therebetween.

The heat exchange unit **104** includes heat exchange tubes communicated between the water inlet part **111** and the water outlet part **112**. The heat exchange tubes surround the flue gas inlet channel **103** below a shielding plate **160**, and heat exchange flues are formed between the tubes. In order to improve the heat exchange efficiency, the water inlet part **111** is disposed close to a bottom of the heat exchanger housing **110**, and the water outlet part **112** is disposed close to a top of the heat exchanger housing **110**. An overall flow direction of the water in the heat exchange unit **104** is from bottom to top, and an overall flow direction of the flue gas is from the lower flue gas inlet **101** to the upper flue gas outlet **102**. The overall flow direction of the water is substantially the same as that of the flue gas, so that the low-temperature flue gas transferred from the primary heat exchanger **200** can fully exchange heat with the water.

As illustrated in FIGS. **2** and **5**, the water inlet part **111** and the water outlet part **112** may be water collecting boxes on the sidewall of the heat exchanger housing **110**, and the water collecting boxes are provided with connecting ports **1121** and **1111**. The water collecting box used as the water outlet part **112** and the water collecting box used as the water inlet part **111** are located on the same sidewall of the housing **110** and arranged at different vertical positions. A water outlet tube **150** in communication with the primary heat exchanger **200** is communicated with the water outlet part **112**, and a water inlet tube for inputting cold water is communicated with the water inlet part **111**.

The heat exchange unit **104** is located between the flue gas inlet channel **103** and the sidewall flue gas exhaust channel **105**. The flue gas outlet **102** is located above the heat exchange unit **104**. Specifically, the flue gas outlet **102** is located above the shielding plate **160**. In which, a top flue gas circulation space is provided between the shielding plate **160** and a top wall of the heat exchanger housing **110**, and the flue gas from the sidewall flue gas exhaust channel **105** flows upward into the top flue gas circulation space, then flows horizontally to converge at the middle, and is discharged to the outside from the flue gas outlet **102** at the center.

In order to avoid unsmooth discharge of flue gas caused by the collision between the flue gases from both sides, a spacing protrusion **115** is disposed between the shielding plate **160** and the top wall of the heat exchanger housing **110** to partition the area above the shielding plate **160** so as to form top flow channels communicated with the flue gas outlet **102** respectively.

As viewed from FIG. **3**, the spacing protrusion **115** is located on the shielding plate **160**, and extends in a front-rear horizontal direction (a direction perpendicular to the paper) to partition the top flue gas circulation space so as to form left and right top flow channels **1151** and **1152**. The spacing protrusion **115** is further provided with diversion slopes to divert the flue gas to the flue gas outlet **102**. The

whole flue gas on the left side (i.e., the flue gas exhausted from about one half of the sidewall flue gas exhaust channel **105**) enters the left top flow channel **1151**, and may be diverted by the spacing protrusion **115** to the flue gas outlet **102** when flowing to the spacing protrusion **115**, and then discharged from the flue gas outlet **102**. Similar to the left top flow channel, the whole flue gas on the right side (i.e., the flue gas exhausted from the other half of the sidewall flue gas exhaust channel **105**) enters the right top flow channel **1152** and is diverted by the spacing protrusion **115** to the flue gas outlet **102** for discharge.

In this embodiment, the flue gas inlet **101** is located on a bottom plate **170** of the heat exchanger housing **110**, and the flue gas outlet **102** is located on a top plate of the heat exchanger housing **110**. On the whole, the flue gas inlet **101** and the flue gas outlet **102** are substantially aligned in a vertical direction, the flue gas inlet **101** is substantially located at a central position of the bottom plate **170** of the heat exchanger housing **110**, and the flue gas outlet **102** is substantially located at a central position of the top plate of the heat exchanger housing **110**.

An end of the heat exchange unit **104** close to the flue gas outlet **102** is shielded and blocked by the shielding plate **160**. The flue gas inlet channel **103** extends from the flue gas inlet **101** to the shielding plate **160**. The shielding plate **160** shields an upper end of the heat exchange unit **104** to prevent the flue gas from flowing directly upwards through the heat exchange unit **104** into the flue gas outlet **102**.

In this embodiment, in order to improve the heat exchange efficiency, the heat exchange unit **104** includes a plurality of heat exchange tube layers **1041** stacked in a vertical direction. As illustrated in FIG. 6, the heat exchange tube layers **1041** include a plurality of heat exchange tube rings nested with one another and having surrounded areas gradually decreasing from outside to inside in a radial direction. The heat exchange flues include spacing flues **1045** located between adjacent two of the heat exchange tube layers **1041** in an up-down direction. The spacing flue **1045** is substantially a horizontal flue, and may also be slightly inclined. Among the plurality of heat exchange tube rings, two heat exchange tube rings may be formed by bending the same heat exchange tube or different heat exchange tubes.

In the same heat exchange tube layer **1041**, a plurality of heat exchange tube rings are located at the same height and nested one within another, so that the interlayer spacing gap **1045** between adjacent two of the heat exchange tube layers **1041** in the up-down direction forms a heat exchange flue communicating the sidewall flue gas channel with the flue gas inlet channel **103**. The heat exchange tube layer **1041** may be wholly placed horizontally, or may be slightly inclined.

The heat exchange tube rings of different heat exchange tube layers **1041** are aligned one by one in the vertical direction (i.e., the flue gas flow direction of the flue gas inlet channel **103** or the extension direction of the flue gas inlet channel **103**). That is, the heat exchange tube rings with the same internal area are aligned in the up-down direction. The heat exchange tube rings with the same surrounded area of the different heat exchange tube layers **1041** are stacked in the vertical direction to form a tube ring stack **1040**, i.e., the heat exchange tube rings with the same internal area are stacked in the vertical direction to form the tube ring stack **1040**. The heat exchange flue further includes a vertical flue located between adjacent two of the tube ring stacks **1040** to communicate a plurality of spacing flues **1045**.

Of course, in other embodiments, the heat exchange tube rings of two heat exchange tube layers **1041** in the up-down

direction can also be staggered, and the heat exchange efficiency can be improved by setting an appropriate spacing between the heat exchange tube layers.

In this embodiment, the (interlayer) vertical spacing between adjacent two of the heat exchange tube layers **1041** is smaller than the (inter-nest) horizontal spacing between adjacent two of the tube ring stacks **1040**. Thus, the flue gas flow resistance is reduced by setting a larger spacing between the tube ring stacks **1040**, thereby lowering the requirement for the fan. Further, in order to improve the heat exchange efficiency, the spacing between adjacent two of the heat exchange tube layers **1041** is greater than 0.68 mm and less than 2.66 mm. The heat exchange efficiency of the condensing heat exchanger **100** is ensured by setting an appropriate spacing between the heat exchange tube layers **1041**.

In order to ensure a gap consistency of the heat exchange unit **104** in the horizontal direction and the vertical direction and then ensure a condensation consistency of the condensing heat exchanger **100**, the heat exchange unit **104** is further provided with spacer bars (**1046**, **1047**) for spacing the heat exchange tube rings. The spacer bars may include a first spacer bar **1047** and a second spacer bar **1046**. Specifically, the heat exchange unit **104** is further provided with the first spacer bar **1047** for spacing adjacent two of the heat exchange tube layers **1041**. The first spacer bar **1047** is disposed between every two of the heat exchange tube layers **1041**. The first spacer bar **1047** between adjacent two of the heat exchange tube layers **1041** substantially extends horizontally or radially for a (interlayer) spacing gap (substantially a horizontal gap/spacing flue **1045**) to form the heat exchange flue. In order to ensure the consistency of the interlayer spacing gaps in the peripheral direction, a plurality of first spacing bars **1047** are distributed at intervals in the peripheral direction in each of the interlayer spacing gaps. Specifically, the first spacer bar **1047** may be a U-shaped structure inserted into adjacent two of the interlayer spacing gaps (substantially the horizontal gap/spacing flue **1045**).

In order to improve the condensing efficiency, the heat exchange unit **104** is further provided with a second spacer bar **1046** for spacing adjacent two of the tube ring stacks **1040** and attached thereto. The second spacer bar **1046** extends in the vertical direction. The second spacer bar **1046** is attached to the tube walls of the tube ring stacks **1040**, which is beneficial for the condensed water on the heat exchange tube to fall down along the second spacer bar **1046**, so as to prevent the condensed water from being always accumulated on the heat exchange tube, thereby improving the condensing efficiency. An inter-nest spacing gap is formed between adjacent two of the tube ring stacks **1040**. As mentioned above, a gap width of the inter-nest spacing gap is greater than that of the interlayer spacing gap, so as to achieve a smaller flue gas flow resistance and lower the requirement for the fan. The second spacer bar **1046** is a U-shaped structure inserted into the inter-nest spacing gap. A plurality of second spacer bars **1046** are provided in the peripheral direction to ensure the width consistency of the inter-nest spacing gaps.

In other words, the heat exchange unit **104** includes a plurality of tube ring stacks **1040** nested layer by layer in the radial direction. The tube ring stack **1040** includes a plurality of heat exchange tube rings having the same surrounded area and being sequentially stacked in the vertical direction. A vertical flue that communicates the spacing flues **1045** in the up-down direction is provided between adjacent two of the tube ring stacks **1040**. The heat exchange tube rings of each of the tube ring stacks **1040** are radially aligned with the heat

exchange tube rings of other tube ring stacks **1040** one by one. The heat exchange tube rings located at the same height and sequentially nested from outside to inside (or from inside to outside, or in the radial direction) form the heat exchange tube layer **1041**. Further, a plurality of heat exchange tube layers **1041** may form a plurality of tube ring stacks **1040**.

As illustrated in FIGS. 7 and 8, the tube ring stack **1040** is formed by one heat exchange tube spirally extending from the water inlet part **111** to the water outlet part **112**. The heat exchange tube layer **1041** is formed by the heat exchange tube rings located at the same height of the different tube ring stacks **1040**. Further, at least two of the tube ring stacks **1040** are connected in parallel between the water inlet part **111** and the water outlet part **112**. In this embodiment, the heat exchange unit **104** includes six tube ring stacks **1040a**, **1040b**, **1040c**, **1040d**, **1040e** and **1040f** connected in parallel, each of which has a lower input end **1048** communicated with the water inlet part **111**, and an upper water outlet end **1049** communicated with the water outlet part **112**.

Of course, in other embodiments, at least two of the tube ring stacks **1040** are connected in series between the water inlet part **111** and the water outlet part **112**. For example, two tube ring stacks **1040** are formed by bending the same heat exchange tube. In some other embodiments, some of the tube ring stacks **1040** may be connected in parallel, and some of the tube ring stacks **1040** may be connected in series.

In a specific embodiment, the heat exchange unit **104** includes a first number of tube ring stacks **1040** connected in parallel and a second number of tube ring stacks **1040** connected in parallel. The first number of tube ring stacks **1040** and the second number of tube ring stacks **1040** are connected in series between the water inlet part **111** and the water outlet part **112**.

In case of six tube ring stacks **1040** each formed by a continuous spiral extension of one heat exchange tube, a communication part (similar to the water collecting box of the water outlet part **112** or the water inlet part **111** in FIG. 5) is formed on the sidewall of the heat exchanger housing **110**, wherein three tube ring stacks **1040** (referred to as the first number of first tube ring stacks **1040**) are connected in parallel between the communication part and the water outlet part **112**, three tube ring stacks **1040** (referred to as the second number of tube ring stacks **1040**) are connected in parallel between the communication part and the water inlet part **111**, and the first number of tube ring stacks **1040** and the second number of tube ring stacks **1040** are communicated through the communication part for serial connection.

In an embodiment where all of the tube ring stacks **1040** are connected in parallel between the water inlet part **111** and the water outlet part **112**, different heat exchange tube rings located in the same heat exchange tube layer **1041** are independent of each other, rather than communicated with each other, and are formed by the heat exchange tube rings located at the same height of the different tube ring stacks **1040**.

In another embodiment, as illustrated in FIG. 9, the heat exchange tube layers **1041a**, **1041b**, **1041c**, **1041d**, **1041e**, **1041f**, **1041g** and **1041h** are formed by one heat exchange tube spirally winding from outside to inside, and are stacked to form a plurality of tube ring stacks **1040**. At least two of the heat exchange tube layers **1041** are connected in parallel between the water inlet part **111** and the water outlet part **112**, and/or at least two of the heat exchange tubes are connected in series between the water inlet part **111** and the water outlet part **112**.

As illustrated in FIG. 9, in an embodiment where all of the heat exchange tube layers **1041** are connected in parallel between the water inlet part **111** and the water outlet part **112**, different heat exchange tube rings of each of the tube ring stacks **1040** are independent of each other, rather than communicated with each other. Each of the heat exchange tube layers **1041** spirally extends inward from a water inlet end **1048'**, a water outlet end **1049'** is inside the heat exchange tube layer **1041** and converges upward to be uniformly communicated with the water outlet part **112**, and the water inlet end of each of the heat exchange tube layers **1041** extends leftward to reach the water inlet part **111** on the sidewall of the heat exchanger housing **110**.

In this embodiment, the bottom of the heat exchanger housing **110** is further provided with a condensed water outlet **113**. In order to collect and discharge the condensed water, the condensed water outlet **113** is located at a lowest position at the bottom of the heat exchanger housing **110**. The condensed water outlet **113** is located on the bottom plate **170** of the heat exchange housing **110** and at one side (the right side in the drawing) of the heat exchange unit **104**. The condensed water outlet **113** collects the converged condensed water and discharges the condensed water out of the condensing heat exchanger **100**.

As illustrated in FIG. 3, in order to improve the condensing efficiency and facilitate the collection of the condensed water, the bottom of the heat exchanger housing **110** has a bearing surface (an inner surface/upper surface of the bottom plate **170**) for bearing the heat exchange unit **104** and wholly inclined to the condensed water outlet **113** from a side away from the condensed water outlet **113**. The height of each of the heat exchange tube layers **1041** (heat exchange units **104**) gradually decreases from the side away from the condensed water outlet **113** to a side close to the condensed water outlet **113**. Accordingly, the heat exchange unit **104** is wholly inclined to a side where the condensed water outlet **113** is located.

In FIG. 3, the condensed water outlet **113** is located on the right side of the heat exchange unit **104**, the bottom plate **170** and the heat exchange unit **104** are inclined from left to right with the height gradually decreased, and the heat exchange tube layer **1041** (heat exchange unit **104**) is inclined so that the condensed water on the heat exchange tube flows to the (vertical or horizontal) spacer bar, then is diverted by the spacer bar to the inclined bottom surface of the heat exchanger housing **110**, and is diverted by the inclined bottom surface to the condensed water outlet **113**, which promotes to discharge all the condensed water inside the heat exchanger, thereby improving the condensing efficiency and the heat exchange efficiency.

In this embodiment, the bottom plate **170** of the heat exchanger housing **110** forms the bearing surface by stamping. The bottom plate **170** of the heat exchanger housing **110** is an integrally stamped structure. The bottom plate **170** of the heat exchanger housing **110** is formed by stamping as a plate for receiving the condensed water, so that a sheet metal welding position on the bottom plate **170** can be lowered and the water leakage caused by poor welding can be prevented. The flue gas outlet **102** is formed on the bottom plate **170** and has an inward flange structure, which extends into the flue gas inlet channel **103**.

Further, the inclination angles of the bearing surface and the heat exchange unit **104** (heat exchange tube layer **1041**) with respect to the horizontal plane are within 10 degrees. Preferably, the inclination angles of the bearing surface and the heat exchange unit **104** (heat exchange tube layer **1041**) with respect to the horizontal plane is within 5 degrees.

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As illustrated in FIG. 1, an embodiment of the present disclosure further provides a water heater, which is a gas water heater, specifically including: a primary heat exchanger 200; and the condensing heat exchanger 100 according to any of the above embodiments. The condensing heat exchanger 100 is above the primary heat exchanger 200 which may be a finned tube heat exchanger. The heat exchange tubes of the primary heat exchanger 200 are communicated with the heat exchange unit 104 of the condensing heat exchanger 100, and specifically, the primary heat exchanger 200 is communicated downstream of the condensing heat exchanger 100.

It should be noted that the combustion part, the primary heat exchanger 200 part and other parts (such as a pipeline connection part, a control part, etc.) of the water heater according to this embodiment may adopt any suitable existing configuration. In order to clearly and concisely explain the technical solution according to this embodiment, the description of the above parts will not be repeated here, and the drawings have been simplified accordingly. However, it should be understood that the scope of this embodiment is not limited thereto.

Any numerical value cited herein includes all lower and upper values that increase by one unit from a lower limiting value to an upper limiting value, as long as there is an interval of at least two units between any lower value and any higher value. For example, if it is stated that the number of certain component(s) or a value of a process variable (e.g., temperature, pressure, time, etc.) is 1 to 90, preferably 20 to 80, and more preferably 30 to 70, the purpose is to illustrate that the specification also explicitly lists values such as 15 to 85, 22 to 68, 43 to 51 and 30 to 32. For a value less than 1, it is appropriate to consider a unit as 0.0001, 0.001, 0.01 or 0.1. These are only examples that are intended to be clearly expressed, and it can be considered that any possible combination of the numerical values listed between a lowest value and a highest value is explicitly set forth in the specification in a similar manner.

Unless otherwise specified, any range includes endpoints and all numerical values between the endpoints. 'About' or 'approximately' used along with a range is suitable for both endpoints of the range. Thus, 'about 20 to 30' is intended to cover 'about 20 to about 30', including at least the indicated endpoints.

All articles and references disclosed, including patent applications and publications, are incorporated herein by reference for all purposes. The phrase 'substantially composed of . . . ' describing a combination should include the determined elements, compositions, components or steps, and any other element, composition, component, or step which substantively does not affect the basic novel features of the combination. When the term 'comprise' or 'include' is used to describe the combination of elements, compositions, components or steps herein, embodiments substantially consisting of the elements, compositions, components or steps are also contemplated. Herein the term 'may' is used to indicate that any described attribute covered by 'may' is optional.

Multiple elements, compositions, components, or steps can be provided by a single integrated element, composition, component or step. Alternatively, a single integrated element, composition, component or step may be divided into multiple separate elements, compositions, components or steps. The disclosure 'a/an' or 'one' used to describe the elements, compositions, components or steps is not intended to exclude other elements, compositions, components or steps.

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It should be understood that the above description is for illustration and not for limitation. By reading the above description, many embodiments and many applications other than the provided examples will be obvious to those skilled in the art. Therefore, the scope of the present teaching should be determined not with reference to the above description, but with reference to the appended claims along with the full range of equivalents to which these claims are entitled. For comprehensive purposes, all articles and references, including the disclosures of the patent applications and announcements, are hereby incorporated by reference. Omitting any aspect of the subject matter disclosed herein in the fore-mentioned claims is not intended to give up the content of the subject matter, nor should it be considered that the inventor has not considered the subject matter as a part of the disclosed subject matter of the invention.

What is claimed is:

1. A condensing heat exchanger, comprising:
a heat exchanger housing having a flue gas inlet and a flue gas outlet; and
a heat exchange unit located within the heat exchanger housing,

wherein a sidewall flue gas exhaust channel communicated with the flue gas outlet is provided between the heat exchange unit and an inner sidewall of the heat exchanger housing, the heat exchange unit surrounds a flue gas inlet channel communicated with the flue gas inlet, the sidewall flue gas exhaust channel surrounds the heat exchange unit, and the heat exchange unit is internally provided with a heat exchange flue which communicates the flue gas inlet channel with the sidewall flue gas exhaust channel, and

wherein the heat exchange unit comprises a plurality of heat exchange tube layers stacked in a vertical direction, the heat exchange tube layers comprise a plurality of heat exchange tube rings nested with one another and having surrounded areas gradually decreasing from outside to inside in a radial direction, and the heat exchange flue comprises a spacing flue located between adjacent two of the heat exchange tube layers in an up-down direction.

2. The condensing heat exchanger according to claim 1, wherein the heat exchange tube rings having the same surrounded area of the different heat exchange tube layers are stacked in the vertical direction to form a tube ring stack.

3. The condensing heat exchanger according to claim 2, wherein the heat exchanger housing is provided with a water inlet part and a water outlet part, the tube ring stack is formed by one heat exchange tube spirally extending from the water inlet to the water outlet, and the heat exchange tube layer is formed by the heat exchange tube rings located at the same height of the different tube ring stacks.

4. The condensing heat exchanger according to claim 3, wherein at least two of the tube ring stacks are connected in parallel between the water inlet part and the water outlet part.

5. The condensing heat exchanger according to claim 3, wherein the heat exchanger housing is provided with a water inlet part and a water outlet part, and at least two of the tube ring stacks are connected in series between the water inlet and the water outlet.

6. The condensing heat exchanger according to claim 3, wherein the heat exchange unit comprises a first number of tube ring stacks connected in parallel and a second number of tube ring stacks connected in parallel, and the first number

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of tube ring stacks and the second number of tube ring stacks are connected in series between the water inlet part and the water outlet part.

7. The condensing heat exchanger according to claim 2, wherein a vertical spacing between adjacent two of the heat exchange tube layers is less than a horizontal spacing between adjacent two of the tube ring stacks.

8. The condensing heat exchanger according to claim 2, wherein the heat exchange unit is further provided with a spacer bar for spacing adjacent two of the tube ring stacks and attached thereto.

9. The condensing heat exchanger according to claim 1, wherein the heat exchange tube layer is formed by one heat exchange tube spirally winding from outside to inside.

10. The condensing heat exchanger according to claim 9, wherein the heat exchanger housing is provided with a water inlet part and a water outlet part, at least two of the heat exchange tube layers are connected in parallel between the water inlet part and the water outlet part, and/or at least two of the heat exchange tubes are connected in series between the water inlet part and the water outlet part.

11. The condensing heat exchanger according to claim 1, wherein a vertical spacing between adjacent two of the heat exchange tube layers is greater than 0.68 mm and less than 2.66 mm.

12. The condensing heat exchanger according to claim 1, wherein the heat exchange unit is further provided with a spacer bar for spacing adjacent two of the heat exchange tube layers.

13. The condensing heat exchanger according to claim 1, wherein a bottom of the heat exchanger housing is further provided with a condensed water outlet located at a lowest position at the bottom of the heat exchanger housing.

14. The condensing heat exchanger according to claim 13, wherein the bottom of the heat exchanger housing has a bearing surface for bearing the heat exchange unit and wholly inclined to the condensed water outlet from a side away from the condensed water outlet.

15. The condensing heat exchanger according to claim 14, wherein a bottom plate of the heat exchanger housing forms the bearing surface by stamping.

16. The condensing heat exchanger according to claim 14, wherein an inclination angle of the bearing surface with respect to a horizontal plane is within 10 degrees.

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17. The condensing heat exchanger according to claim 13, wherein a height of the heat exchange unit or at least one of the heat exchange tube layers gradually decreases from a side away from the condensed water outlet to a side close to the condensed water outlet.

18. A condensing heat exchanger, comprising:

a heat exchanger housing having a flue gas inlet and a flue gas outlet; and

a heat exchange unit located within the heat exchanger housing,

wherein a sidewall flue gas exhaust channel communicated with the flue gas outlet is provided between the heat exchange unit and an inner sidewall of the heat exchanger housing, the heat exchange unit surrounds a flue gas inlet channel communicated with the flue gas inlet, the sidewall flue gas exhaust channel surrounds the heat exchange unit, and the heat exchange unit is internally provided with a heat exchange flue which communicates the flue gas inlet channel with the sidewall flue gas exhaust channel, and

wherein the heat exchanger housing is provided with a water inlet part and a water outlet part, and the heat exchange unit comprises a heat exchange tube communicated between the water inlet part and the water outlet part.

19. The condensing heat exchanger according to claim 18, wherein an end of the heat exchange unit close to the flue gas outlet is shielded and blocked by a shielding plate, and the flue gas inlet channel extends from the flue gas inlet to the shielding plate.

20. The condensing heat exchanger according to claim 19, wherein a spacing protrusion is disposed between the shielding plate and a top wall of the heat exchanger housing to partition an area above the shielding plate to form top flow channels communicated with the flue gas outlet, respectively.

21. The condensing heat exchanger according to claim 18, wherein the water inlet part is disposed close to a bottom of the heat exchanger housing, and the water outlet part is disposed close to a top of the heat exchanger housing.

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